Azure Cloud Interview Guide

30 Questions — Interviewer Level | English Answer + Hindi/Hinglish Explanation Landing Zone • Networking • VM • Terraform

SECTION 1 — Azure Landing Zone

Q **1. What is a Landing Zone and why does an organization need it?**

A Pre-configured Azure environment with networking, security, identity & governance already set up. Every team gets a compliant, ready-to-use environment — no need to build from scratch. Prevents inconsistent configs, insecure setups, and IP conflicts across teams.

HI *Ek taiyar ghar jaisa — neev, bijli, paani sab pehle se set. Har team seedha kaam shuru kare, bina sab khud banaye. Bina Landing Zone ke 50 teams = 50 alag messy setups.*

■ *Interview Tip: Mention CAF (Cloud Adoption Framework) and 'governance at scale' — interviewer will be impressed.*

Q **2. What are the major pillars of a Landing Zone?**

A CAF defines 8 critical design areas: Identity & Access, Network Topology, Security & Governance, Management & Monitoring, Business Continuity, Subscription Organization, Platform Automation, and Enrollment/Tenants.

HI *Yaad rakho: IN-SMG-BP-A — Identity, Networking, Security, Management, Governance, Business continuity, Platform automation, Azure AD. Ek bhi weak ho toh poora setup insecure.*

■ *Interview Tip: Say 'CAF 8 critical design areas' — shows real implementation knowledge.*

Q **3. What information would you collect before designing a Landing Zone?**

A Business: teams, regulatory needs (PCI/HIPAA). Technical: on-prem IP ranges (avoid overlap!), hybrid identity, connectivity (VPN/ExpressRoute). Operational: tagging, cost ownership, CCoE model. Growth: expected subscription count in 1–3 years.

HI *Sabse badi galti — on-prem IP range check na karna. Agar on-prem 10.0.0.0/8 use karta hai aur Azure bhi wahi le le — VPN pe routing blast. Pehle IPAM check karo!*

■ *Interview Tip: Mention 'IP overlap gotcha' and 'compliance requirements' — shows real-world thinking.*

Q **4. How would you design a Management Group and Subscription hierarchy?**

A Tenant Root → Platform MG (Connectivity, Management, Identity subs) → Landing Zones MG (Corp + Online) → Sandbox MG → Decommissioned MG. Azure Policy top-down inherits — apply at MG level, everything below auto-complies.

HI *Socho company structure: Head Office (Root) → Departments (MGs) → Teams (Subscriptions). Connectivity/Management/Identity alag subscriptions me isliye — blast radius isolation aur separate billing.*

■ *Interview Tip: 'Why separate Connectivity subscription?' — answer: blast radius + independent lifecycle.*

Q **5. What is Enterprise Scale Landing Zone?**

A Microsoft's opinionated CAF reference implementation — 100+ Azure Policies, Hub-Spoke networking, Log Analytics, Defender for Cloud — all pre-configured. GitHub: Azure/Enterprise-Scale. Supports Bicep & Terraform. Key concept: 'Subscription Vending' — new subscriptions auto-configured via pipeline.

HI *IKEA kit jaisa — ghar khud design karne ki jagah ready blueprint lo. Lekin blindly deploy mat karo — organization ke custom needs ke liye customize zaroor karo.*

■ *Interview Tip: Say 'Subscription Vending Machine' concept — senior/principal level knowledge.*

SECTION 2 — Azure Networking

Q **6. What is CIDR and how do you choose a CIDR range in production?**

A /16 = 65,536 IPs, /24 = 256 IPs, /28 = 16 IPs (Azure minimum subnet). Production: Hub /22, Spokes /23–/24. Always RFC 1918 (10.x, 172.16–31.x, 192.168.x). Note: Azure reserves 5 IPs per subnet (.0, .1, .2, .3, .255).

HI */28 subnet liya toh sirf 11 usable IPs milenge — 5 Azure apne rakhta hai. Baad me expand karna ho toh existing resources affect hote hain. Isliye pehle se bada lo!*

■ *Interview Tip: Say '5 reserved IPs per subnet' and 'on-prem overlap check via IPAM' — real-world experience.*

Q **7. What factors do you consider when choosing a CIDR range?**

A IPAM tool se allocated range lo (NetBox/Infoblox). Peered VNets must be non-overlapping. Subnet delegation needs (AKS, App Service Env). SNAT port exhaustion risk. ExpressRoute route table limit (4,000 max). Summarizable ranges preferred for on-prem advertisement.

HI *Peering ke baad overlap mila — toh peering delete karo, IP change karo, dobara banao. Production me ye ek hafte ka kaam hai. Pehle sahi plan karo.*

■ *Interview Tip: Say 'SNAT port exhaustion' and 'ExpressRoute 4000 route limit' — shows network depth.*

Q **8. What is the difference between VNet and Subnet?**

A VNet = Layer 3 boundary, region-scoped, full IP space. Subnet = VNet partition — NSG, Route Table, Service Delegation all at subnet level. Special subnets: GatewaySubnet (exact name mandatory, no NSG), AzureFirewallSubnet (/26 min), AzureBastionSubnet (/26 min).

HI *VNet = city, Subnet = mohalla. GatewaySubnet ka naam exact 'GatewaySubnet' hona chahiye — ek bhi letter wrong toh VPN Gateway deploy nahi hoga. Common interview gotcha hai ye.*

■ *Interview Tip: 'GatewaySubnet exact naming' and 'no NSG on GatewaySubnet' — common gotcha interviewers ask.*

Q **9. What is Hub-Spoke architecture and why is it used?**

A Hub = shared services VNet (Firewall, VPN GW, Bastion, DNS). Spokes = workload VNets. Force Spoke-to-Spoke traffic through Hub Firewall via UDR — east-west inspection. Alternative: Azure Virtual WAN (Microsoft-managed, better at scale).

HI *Spoke-to-Spoke direct nahi jaata — UDR se 0.0.0.0/0 → Azure Firewall point karo. Ye step bhool gaye toh VMs directly baat kar sakti hain — security bypass! Azure Firewall ~$1.25/hr extra cost.*

■ *Interview Tip: 'Spoke-to-Spoke via UDR force tunneling' — always explain this, shows architecture depth.*

Q

10. What is the difference between VNet Peering and VPN Gateway?

A

Peering: Azure backbone, ultra-low latency, no encryption needed, non-transitive by default. VPN Gateway: IPSec encrypted, for on-prem or cross-tenant. Gateway Transit: Spokes use hub GW (UseRemoteGateway=true). ExpressRoute: Private dedicated circuit, no internet.

HI

Non-transitive = A-B peered, B-C peered, lekin A-C directly baat nahi kar sakti. Hub me NVA/Firewall lagao aur UDR se route karo — tabhi transitivity milti hai. Ye common VNet architecture trap hai.

■ *Interview Tip: 'Non-transitive peering' and 'Gateway Transit flag' — comes up in senior networking questions.*

Q

11. What is the difference between NSG and Azure Firewall?

A

NSG: Layer 4, free, subnet/NIC level, 5-tuple rules, no FQDN/TLS inspection. Azure Firewall Premium: Layer 7 NGFW, FQDN filtering, TLS inspection, IDPS, Threat Intelligence — ~$1.25/hr + data charges. Production: NSG every subnet (defense-in-depth) + Azure Firewall for central inspection.

HI

NSG = ghar ka tala (basic). Azure Firewall = CCTV + security guard + visitor log (advanced). Dono saath use karo — NSG pehli line of defense, Firewall centralized inspection. Sirf NSG enterprise ke liye enough nahi.

■ *Interview Tip: 'Defense-in-depth — use both' and 'IDPS is a Premium feature' — shows architecture thinking.*

Q

12. What is a Route Table (UDR)?

A

Overrides Azure's default system routes. Common: 0.0.0.0/0 → Azure Firewall (force all internet traffic). Must disable BGP Route Propagation — otherwise on-prem routes can override UDR. Troubleshoot via NIC → Effective Routes in portal.

HI

UDR lagaya lekin BGP propagation disable nahi kiya — on-prem routes ne UDR override kar diya aur traffic firewall bypass ho gaya. Production incident! Always BGP propagation disable karo saath me.

■ *Interview Tip: 'BGP propagation disable with UDR' and 'Effective Routes for troubleshooting' — real ops experience.*

Q

13. What is the difference between Private Endpoint and Service Endpoint?

A

Service Endpoint: Public endpoint remains, traffic via Microsoft backbone, free, data exfiltration possible. Private Endpoint: Private IP in your VNet, NIC created, fully private, prevents data exfiltration. Critical: Link Private DNS Zone — otherwise public IP resolves and traffic goes over internet!

HI

Private Endpoint lagaya lekin Private DNS Zone link nahi ki — traffic public internet se gaya. Security bypass! DNS integration miss karna #1 mistake hai. Ye production me real incident hota hai.

■ *Interview Tip: 'DNS integration gotcha' and 'data exfiltration prevention' — clearly explain, interviewer wants to hear this.*

Q

14. What is the difference between Application Gateway and Load Balancer?

A

ALB: Layer 4 (TCP/UDP), ultra-fast, no HTTP awareness, Standard SKU zone-redundant. App Gateway v2: Layer 7 (HTTP/HTTPS), URL routing, SSL offload, WAF (OWASP rules), multi-site hosting, autoscaling. Azure Front Door = global CDN + AGW. Traffic Manager = DNS-based global LB.

HI

LB = traffic police (left ya right). App Gateway = smart concierge (URL path dekh ke room assign karta hai). Web app ke liye AGW + WAF, raw TCP ke liye LB. Global chahiye? Front Door use karo.

■ *Interview Tip: Say 'AGW v2 autoscaling' and 'Front Door vs AGW — global vs regional' comparison.*

Q **15. Why do we use Azure Bastion?**

A VMs with public IP + open RDP/SSH = huge attack surface. Bastion: Browser-based RDP/SSH from Azure portal, no public IP on VM, deployed in AzureBastionSubnet (/26 min). Standard SKU: native RDP client, file transfer. Combine with JIT Access for on-demand port opening.

HI *Bastion + JIT + PIM = tripled security. Bastion se connect, JIT se 3-hour window open, PIM se admin role just-in-time. Ye trifecta production me use karo.*

■ *Interview Tip: 'Bastion Standard vs Basic' and 'JIT as complement' — shows layered security knowledge.*

Q **16. How does DNS resolution work in Azure?**

A Azure DNS default: 168.63.129.16. Private DNS Zones: internal names, auto-registration. Azure DNS Private Resolver: Conditional forwarding on-prem ↔ Azure (replaces DNS forwarder VMs). Custom DNS: set in VNet settings. Split-horizon DNS: same domain, different IPs inside/outside.

HI *Private Endpoint lagao + Private DNS Zone link mat karo — public IP resolve hoga, traffic internet jaega. DNS Private Resolver ne DNS forwarder VMs ko replace kar diya — ab IaaS VM nahi chahiye sirf DNS ke liye.*

■ *Interview Tip: Say 'DNS Private Resolver' (modern approach) and 'Private Endpoint DNS gotcha' — advanced DNS knowledge.*

Q **17. How would you design a multi-region network architecture?**

A Active-Active: Hub-Spoke per region, Global VNet Peering between hubs, Azure Front Door (HTTP global LB + WAF), Traffic Manager (DNS-based non-HTTP). Data sync: Cosmos DB multi-region writes, SQL geo-replication. Active-Passive for DR. Define RTO/RPO first — that drives architecture.

HI *Pehle poochho: RTO/RPO kya hai? 15 min RTO = Active-Active chahiye. 4 hour RTO = Active-Passive theek hai. Bina ye jaane architecture design mat karo — interviewer yahi bolega.*

■ *Interview Tip: 'RTO/RPO drives architecture choice' and 'Front Door vs Traffic Manager' — design thinking.*

SECTION 3 — Azure Virtual Machines

Q **18. What components are required to create an Azure VM?**

A Required: Resource Group, Region, VM Size (SKU), OS Image, OS Disk, VNet+Subnet, NIC, Authentication (SSH key preferred). Production must-haves: No public IP (use Bastion), Availability Zone, Managed Identity, Boot Diagnostics, Azure Monitor Agent, separate Data Disks.

HI *Managed Identity = application ke liye bina password ke Azure services access. Key Vault me secret daalo, VM Managed Identity se read karo — no hardcoded credentials. Ye production security ka core hai.*

■ *Interview Tip: Say 'Managed Identity' and 'no public IP + Bastion' — security-conscious architect answer.*

Q **19. What is the difference between Availability Set and Availability Zone?**

A AS: Same datacenter, different racks (FDs/UDs), 99.95% SLA, free. AZ: Physically separate datacenters same region, 99.99% SLA, inter-zone data transfer costs. Use AZ for production-critical. Use AS for services that don't support AZ.

HI *99.95% vs 99.99% = sirf 0.04% difference lagta hai? Nahi! 99.95% = 4.38 hrs downtime/year. 99.99% = sirf 52 min/year. Production SLA ke liye ye fark bahut bada hai.*

■ *Interview Tip: '99.95% vs 99.99% downtime hours' — SLA-aware architect answer.*

Q **20. What are Fault Domains and Update Domains?**

A FD: Physical rack = same power + switch. AS has 2–3 FDs. FD0 down = only FD0 VMs affected. UD: Logical grouping for Azure maintenance. Default 5 UDs, max 20. Azure restarts one UD, waits 30 min, then next. Example: 6 VMs, 2FD, 5UD — FD0 fails, 3 VMs down, 3 running.

HI *Interview me concrete example bolna best hai: '6 VMs, 2 FDs, 5 UDs — VM0 FD0/UD0, VM1 FD1/UD1... FD0 fails toh VM0,VM2,VM4 down, baaki 3 chalu.' Ye sunke interviewer samjhega tumne actually use kiya hai.*

■ *Interview Tip: Give the concrete 6-VM example — shows hands-on experience, not just theory.*

Q **21. How are Private IP and Public IP assigned to a VM?**

A Private IP: Via NIC from Azure DHCP (168.63.129.16). Dynamic changes on stop/deallocate (not restart). Static recommended for servers. Public IP: Separate resource. Standard SKU = zone-redundant, secure by default (inbound blocked). Basic = open by default — avoid in prod. IMDS: http://169.254.169.254 for VM self-metadata.

HI *Standard Public IP = secure by default — sab inbound blocked. Basic = open by default. Production me hamesha Standard, lekin better hai public IP hi na do — Bastion use karo.*

■ *Interview Tip: 'Standard vs Basic Public IP security difference' and 'IMDS endpoint' — shows depth.*

Q **22. What are the disk types available for Azure VMs?**

A Standard HDD: <500 IOPS, archive/backup only. Standard SSD: dev/test. Premium SSD: production (20K IOPS, 900 MB/s). Premium SSD v2: flexible IOPS/throughput, sub-ms latency, databases. Ultra Disk: 160K IOPS, 2 GB/s, SAP HANA, top-tier SQL. Shared Disk: for clustering.

HI *Ultra Disk ka IOPS on-the-fly change hota hai — resize nahi karna. Premium SSD v2 bhi flexible. Cost: Premium SSD P30 1TB ~$135/mo, Ultra ~$300/mo + IOPS charges alag.*

■ *Interview Tip: 'Ultra Disk IOPS configurable without resize' and 'VM throughput ceiling check' — performance tuning knowledge.*

Q **23. What is the difference between OS Disk, Data Disk, and Temporary Disk?**

A OS Disk: OS + pagefile, persistent, ReadWrite caching OK. Ephemeral OS Disk: cache/temp-backed, ultra-fast boot, stateless VMs (AKS nodes). Data Disk: App/DB data, persistent. Cache: ReadOnly for SQL data files, None for SQL log files. Temp Disk: Local SSD, D: (Win)/mnt (Linux), DATA LOST on stop/deallocate/resize.

HI *Temp disk pe database rakha ek engineer ne — VM resize kiya aur saara data gaya. Production incident! D: drive = hotel ka room (checkout pe sab saaf). Important data hamesha Data Disk pe.*

■ *Interview Tip: 'Ephemeral OS Disk for AKS' and 'SQL disk caching strategy' — real workload knowledge.*

Q **24. What factors do you consider when selecting a VM size?**

A Workload type: B (burstable/dev), D (general), E (memory/DB), F (compute), L (storage/NoSQL), N (GPU/ML), H (HPC). Factors: vCPUs, RAM, disk IOPS ceiling, network BW, AZ support. Cost: Reserved 1/3yr (72% savings), Spot (90%, fault-tolerant only), Azure Hybrid Benefit (existing licenses).

HI *B-series = CPU credits jama karo idle me, burst karo peak pe. Dev/test ke liye perfect. Prod ke liye B nahi — guaranteed performance chahiye. Azure Advisor regularly dekho for right-sizing recommendations.*

■ *Interview Tip: Say 'B-series bursting', 'Reserved vs Spot trade-offs', 'Azure Advisor right-sizing'.*

Q **25. How would you secure a production VM?**

A Network: No public IP + Bastion + NSG least-privilege + JIT Access. Identity: RBAC minimum, Managed Identity for apps, PIM for admin. Data: ADE (BitLocker/DM-Crypt) + CMK in Key Vault. OS: Azure Update Manager auto-patching, Defender for Endpoint, Guest Configuration policies. Monitor: Defender for Servers P2, Monitor Agent, Boot Diagnostics.

HI *Trifecta yaad rakho: Bastion (connect karo) + JIT (port temporarily kholo) + PIM (admin role temporarily lo). Teeno saath use karo. CMK = Customer Managed Keys — compliance ke liye apni key Key Vault me.*

■ *Interview Tip: 'JIT + Bastion + PIM trifecta' and 'CMK for disk encryption' — senior security architect answer.*

SECTION 4 — Terraform

Q **26. What is a Terraform state file and why is it important?**

A terraform.tfstate = Terraform's source of truth. Stores resource IDs, attributes, dependencies. Used in plan (diff), apply (update), destroy (identify). Key ops: state list, state mv (refactoring), state rm, import. State drift: manual Azure changes = mismatch — fix via terraform refresh or import.

HI *State file delete ho gayi — terraform dobara sab kuch banana chahega! Disaster. State file = Terraform ki yaaddasht. Khoti toh sab bhool jaata hai. Isliye remote backend mandatory hai production me.*

■ *Interview Tip: Say 'state drift and terraform import', 'state mv for module refactoring' — hands-on experience.*

Q **27. Why do we use a Remote Backend?**

A Local state problems: single point of failure, no concurrent access, no history, sensitive values exposed. Azure Storage Backend: Blob Lease = state locking, Blob versioning = history, RBAC = access control, CMK encryption. Best practice: separate storage account, private endpoint, CI/CD SP gets Contributor only.

HI *Blob Lease = database lock jaisa. Ek engineer apply kar raha hai toh doosra wait karta hai. Bina lock ke dono ne saath apply kiya — state corrupt, production down. Remote backend = collaborative Terraform ka aadhaar.*

■ *Interview Tip: 'Blob Lease mechanism for locking' and 'avoid secrets in state — use Key Vault references'.*

Q **28. What is the difference between count and for_each?**

A count: Index-based. Remove middle item → index shift → Terraform destroys + recreates remaining. DANGEROUS in prod. for_each: Key-based (map/set). Remove 'dev' → only dev RG deleted, others untouched. Rule: count only for toggles (count = var.enable ? 1 : 0). Everything else = for_each.

HI *count se 3 RGs banaye: rg-0, rg-1, rg-2. rg-1 delete kiya — Terraform ne rg-2 ko rg-1 me rename kiya = DESTROY + CREATE. Production me ye ek hafte ka incident hai. for_each se banaya hota toh sirf rg-1 delete hoti.*

■ *Interview Tip: 'count index shift = destroy+recreate' real scenario — clear differentiator for interviewers.*

Q **29. What is a Nested Map and when do you use it?**

A Map inside map/object. Used for complex multi-resource configs. Example: spoke_config = { prod = { cidr = '10.1.0.0/22', subnets = { app = { cidr='10.1.0.0/24' }, db = { cidr='10.1.1.0/24' } } } }. Iterate outer with for_each, access inner via each.value.subnets. Use lookup() for safe access with defaults.

HI *Contacts app jaisa: naam > phone > email. Terraform me: environment > VNet > subnet. Ek variable me poora infrastructure define — clean, reusable, DRY. Real use: multi-region, multi-env, multi-subnet deployments.*

■ *Interview Tip: Give a real multi-subnet nested map example — shows you've written complex Terraform, not just tutorials.*

Q **30. How would you design an enterprise Terraform folder structure and reusable modules?**

A Structure: /modules (hub-vnet, spoke-vnet, linux-vm, key-vault) + /environments (dev/staging/prod with main.tf, variables.tf, tfvars, backend.tf) + /platform (management-groups, policies, connectivity). Module rules: single responsibility, Git tag versioning (v1.2.0), validation blocks, outputs exposed, README mandatory. CI/CD: PR → fmt+validate+tflint+plan (comment) → merge → apply (prod needs approval). Advanced: Terragrunt for DRY, Atlantis for GitOps.

HI *Module banaya lekin version pin nahi kiya — kisi ne module update kiya, sab environments me breaking change. Git tag versioning mandatory: ?ref=v2.1.0. Ye production lesson hai jo ek baar seekha toh kabhi nahi bhoolte.*

■ *Interview Tip: 'Module versioning with Git tags', 'CI/CD plan on PR comment', 'Terragrunt for DRY' — principal level answer.*